

Descent rate calculations

Assumptions

Using the fundamentals of physics, we can derive the formula for the descent rate calculations.

Variable	Value	Reason
C_d Container	1.0	Cylindrical shape with mild streamling
C_d Payload	1.5	Random shape - higher drag
Full mass	0.72 ± 0.2 kg	
Mass of container	0.27 kg	
Mass of payload	0.45 kg	
Friction coefficient	0.3	PLA
First parachute	0.35 m in d	
Second parachute	0.247 m in d	
ρ	1.225 kg/m ³	Air density

Descent rate calculations

$$mg = \frac{1}{2}\rho AC_d V^2$$

$$2mg = \rho AC_d V^2$$

$$V^2 = \frac{2mg}{\rho AC_d}$$

$$V = \sqrt{\frac{2mg}{\rho AC_d}}$$

$$v_t = \sqrt{\frac{2mg}{\rho AC_d}}$$

$$A = \pi \left(\frac{d}{2}\right)^2$$

$$v_t = \sqrt{\frac{2 \cdot 0.72 \cdot 9.81}{1.225 \cdot \pi \cdot \left(\frac{0.257}{2}\right)^2 \cdot 1.5}}$$

$$v_t \approx 12.2 \text{ m/s}$$

Using the fundamentals of physics, we can derive the formula for the descent rate calculations.

The cross-sectional area will be the area of the parachute only because its drag coefficient is dominant.

We take the diameter of the parachute 0.257 m (the maximum one) and calculate the terminal velocity.

Descent rate calculations For Container

The area of the parachute will be the same as in calculations relating to Payload+Container.

We take the diameter of the parachute 0.257 m(the maximum one) and calculate the terminal velocity.

$$v_t = \sqrt{\frac{2 \cdot 0.3 \cdot 9.81}{1.225 \cdot 1.5 \cdot 0.257}} = \sqrt{\frac{5.886}{0.4722}} \approx \sqrt{12.46} \approx 3.53 \text{ m/s}$$

Descent rate calculations For Payload

Using the fundamentals of physics, we can derive the formula for the descent rate calculations.

The cross-sectional area will be the area of the parachute only because its drag coefficient is dominant.

We take the diameter of the parachute 0.35 m (the maximum one) and calculate the terminal velocity.

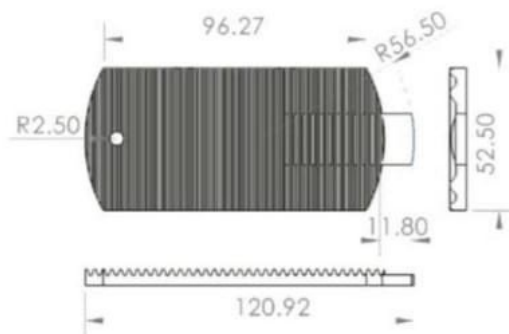
$$A = \pi \left(\frac{0.35}{2} \right)^2 = \pi (0.175)^2 \approx 0.0962$$
$$v_t = \sqrt{\frac{2 \cdot 0.45 \cdot 9.81}{1.225 \cdot 1.5 \cdot 0.0962}} = \sqrt{\frac{8.829}{0.1769}} \approx \sqrt{49.91} \approx 7.06$$

Mechanical subsystem overview

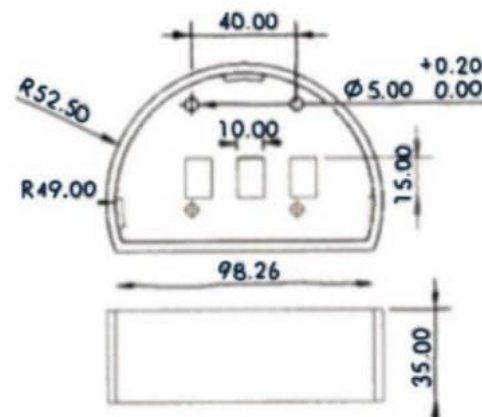
Drawings.

In this section, the team provided sketches and drawings with corresponding dimensions for each element of the mechanical design on order to illustrate and deepen the knowledge into the design. Dimensions can be changed if there is a necessity, but the design in general remains the same

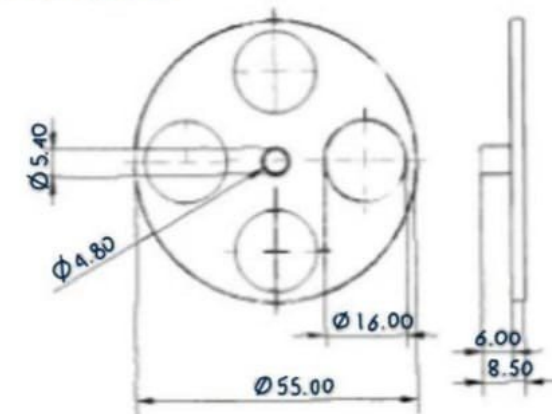
Lid.



Parachute Board.

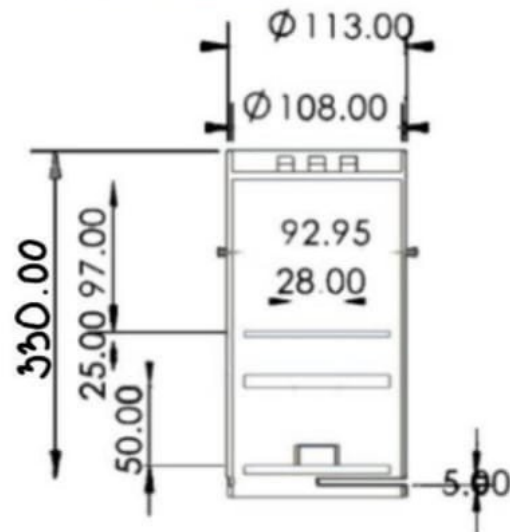


Filter.



Mechanical subsystem overview

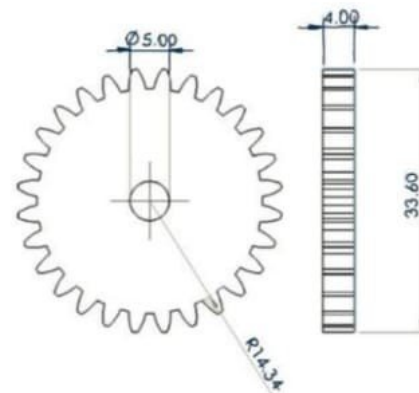
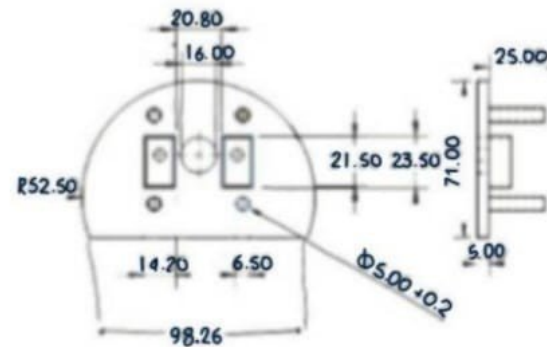
Drawings. Container.



Straw.

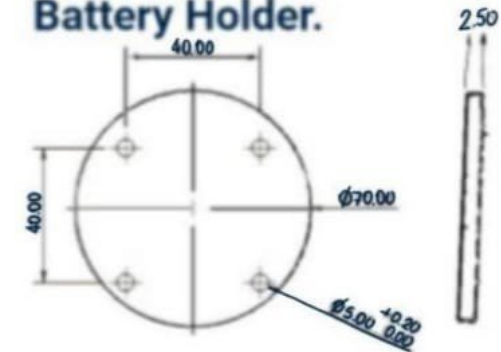


Camera holder.



Pressure angle: 20
Module: 1
Teeth: 29

Battery Holder.

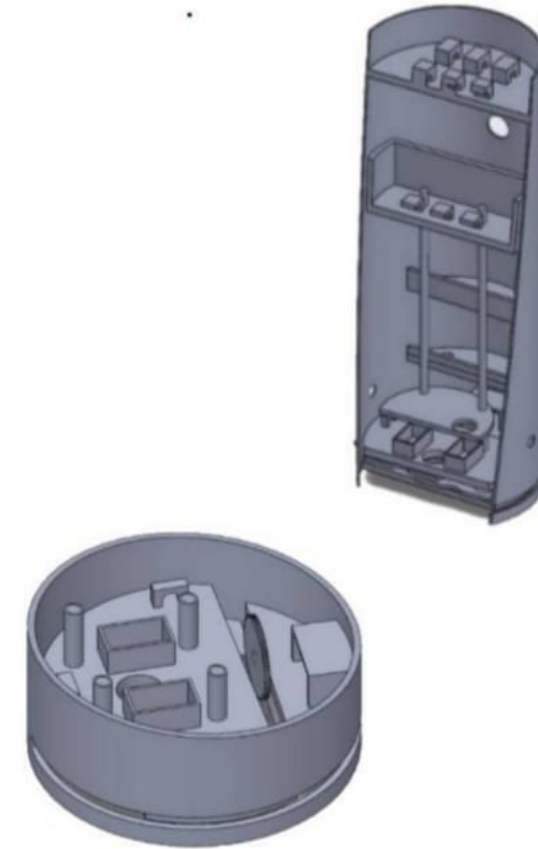


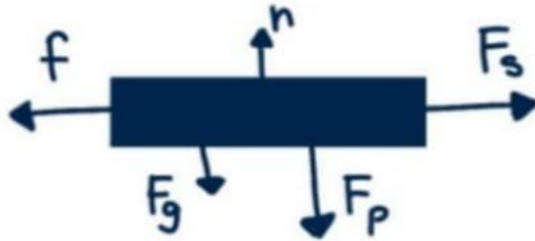
Separation mechanism

The concept of opening mechanism is simple yet requires accurate calculations to estimate not only sustainability of the mechanical design but also to choose appropriate electronics and actuators. The same was considered while designing CanSat satellite. In this case, the team considered using an opening mechanism from the down to ensure that no misgivings will occur during the competition. Every force exerted on the surface was calculated and based on the equations the team concluded using servo motor MG90S. The gear presented on the previous file will be attached to the servo motor, thereby enabling it to rotate clockwise and then counterclockwise to ensure that the lid will come out of its place and let the payload to get out.

Experiment conduction.

The system test can be conducted while holding in air since the conditions in reality and experiment are identical. The same will be done with parachute, but by throwing container and payload at certain altitudes on soft surface in order not to get it damaged.





$$W = m \cdot g = 0.4 \cdot 9.81 = 3.924 \text{ N}$$

$$F_{\text{friction}} = f \cdot W = 0.3 \cdot 3.924 = 1.1772 \text{ N}$$

$$\tau_{\text{friction}} = F_{\text{friction}} \cdot r = 1.1772 \cdot 0.05 = 0.05886 \text{ Nm}$$

$$\tau_{\text{required}} = W \cdot r + \tau_{\text{friction}} = 0.1962 + 0.05886 = 0.25506 \text{ Nm}$$

$$\Delta\tau = \tau_{\text{MG90S}} - \tau_{\text{required}} = 0.31 - 0.25506 = 0.05494 \text{ Nm}$$

$$\frac{\tau_{\text{required}}}{\tau_{\text{MG90S}}} = \frac{0.25506}{0.31} \approx 0.823$$

Lid.

In these calculations, the team decided to estimate whether MG90S is a suitable servo motor for rotating the lid and eventually making the payload to come out. There are three forces exerted on a lid vertically, and two horizontally yet in two different directions.

The team calculated the torque considering the friction which is usually 0.3 for PLA and obtained 0.31 Nm, which exceeds torque of 0.25506 Nm, giving a safety margin of approximately 17.7%. Thus, the MG90S is suitable for the task, though with less margin due to friction which can be neglected if lubricated

$$m = 0.4 \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

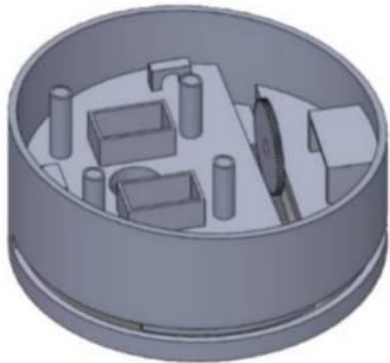
$$r = 0.05 \text{ m}$$

$$\tau_{\text{MG90S}} = 0.31 \text{ Nm}$$

$$f = 0.3$$

Mechanical Filtering module.

Here are two filter holders attached straightly to the servo motor. One of them is longer than another one. The filters are made out of and servo motors will rotate the filter holders and the corresponding image is going to be displayed coloured cast acrylic sheets which are available in every stationary.



Time	Filter	Time	Filter	The Expected Image At the Time of Flight			
6	B	7	G				